ABES ENGINEERING COLLEGE, GHAZIABAD

Department of Mechanical Engineering

FINAL YEAR PROJECT REPORT

ON

DESIGN AND CFD ANALYSIS OF EPPLER E63 UAV PROPELLER FOR QUADCOPTER APPLICATION

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DECLARATION

We hereby declare that this submission is entirely our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person, nor any material that has been accepted, to any substantial extent, for the award of any other degree or diploma at this university or any other institution of higher learning — except where due acknowledgement has been made within the text.

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CERTIFICATE

This is to certify that the project report entitled **"Design and CFD Analysis of Eppler E63 UAV Propeller for Quadcopter Application"**, submitted by **Gyanendra Jha, Jatin Kumar Singh,** and **Dev Kumar** in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in the Department of Mechanical Engineering, ABES Engineering College, Ghaziabad, is a record of the candidates' own work carried out under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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Finally, a word of thanks to our families and friends — for tolerating our erratic schedules, sitting through our rehearsal presentations, and never once complaining about it.

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ABSTRACT

This project documents the complete design chain and aerodynamic investigation of a propeller built around the Eppler E63 low-Reynolds-number airfoil, intended for a 1.2 kg quadcopter UAV. The rotor is two-bladed, 0.25 m in diameter, and was sized to spin at 6400 RPM with a top forward speed of 16 m/s. The E63 profile was chosen because it was purpose-engineered for the Reynolds number window (Re ≈ 60,000–300,000) that governs this class of small rotorcraft — a regime where conventional NACA sections fall apart aerodynamically. Blade Element Momentum Theory (BEMT) with Prandtl tip-loss corrections formed the analytical backbone. Station-by-station calculations fixed the chord and twist distributions, from which hover thrust (5.886 N per rotor, C_T = 0.1082), Figure of Merit (0.80), and cruise propulsive efficiency (80%) were derived. Compared with an equivalent NACA 4412 design, this propeller saves roughly 6.2% in hover power — which translates directly into longer battery endurance in practice. CFD simulations were carried out in ANSYS Fluent 2023 R1 using the Realizable k-epsilon turbulence model with Enhanced Wall Treatment. A polyhedral mesh of approximately 1.8 million cells, with 12-layer prismatic boundary layers targeting y^+ = 2–8 on blade surfaces, was generated in ANSYS Fluent Meshing. Four angles of attack (0°, 4°, 8°, 12°) and three blade materials — carbon fibre reinforced polymer (CFRP), PLA, and Nylon — were investigated. At the design point (4°, Re = 100,000), the E63 reaches a peak C_L/C_D of roughly 63. CFD thrust coefficients agree with BEMT predictions to within 8–12%. Structurally, CFRP stands far ahead of the polymers, showing tip deflection below 0.3 mm versus 2.1 mm for PLA at full operating speed.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AoA	Angle of Attack
BEMT	Blade Element Momentum Theory
CAD	Computer-Aided Design
CFRP	Carbon Fibre Reinforced Polymer
CFD	Computational Fluid Dynamics
C_D	Drag Coefficient
C_L	Lift Coefficient
C_P	Power Coefficient
C_T	Thrust Coefficient
EWT	Enhanced Wall Treatment
FM	Figure of Merit
FSI	Fluid-Structure Interaction
ISA	International Standard Atmosphere
J	Advance Ratio
MRF	Multiple Reference Frame
NACA	National Advisory Committee for Aeronautics
PLA	Polylactic Acid
RANS	Reynolds-Averaged Navier-Stokes
Re	Reynolds Number
RPM	Revolutions Per Minute
UAV	Unmanned Aerial Vehicle
UIUC	University of Illinois at Urbana-Champaign

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Over the past decade or so, small quadcopters have gone from being hobbyist toys to serious working machines. You find them inspecting power lines, mapping crop fields, delivering parcels in controlled trials, and scouting disaster zones where sending a human is too dangerous. The global market for these platforms is now valued well above USD 30 billion, and it keeps growing. That kind of commercial pull brings engineering pressure with it.

Every kilogram of quadcopter has four propellers turning at high speed, and those propellers are arguably the most consequential component in the whole vehicle. They are what actually converts motor shaft torque into the thrust that keeps the machine in the air. Poor propeller design does not just waste energy — it can introduce vibrations that blur camera images, shorten bearing life, and in extreme cases cause structural failure mid-flight. Getting the aerodynamics right, from the outset, matters enormously.

The aerodynamic environment that a small UAV propeller operates in is genuinely unusual. Because the blades are short and spin at moderate speeds, the local Reynolds numbers across the blade span are typically between 20,000 and 150,000. That is a regime where boundary layers behave very differently from the high-Re flow around, say, a commercial aircraft wing. Laminar separation bubbles form easily. Viscous drag is proportionally much higher. Airfoils designed for large aircraft — the NACA four-digit series being the most familiar example — were simply never intended for this range, and their performance degrades badly when forced to work there.

Purpose-designed low-Reynolds-number profiles, particularly the Eppler series developed by Richard Eppler, address these problems directly. The E63 in particular targets the exact Reynolds number window that governs this project's propeller design, making it a natural starting point.

1.2 Problem Statement

Most off-the-shelf quadcopter propellers come from injection-moulding machines, not from optimisation studies. The chord and twist distributions are determined largely by manufacturing convenience rather than aerodynamic theory. The result is propellers that work well enough for casual use but leave a measurable amount of efficiency on the table. Four specific shortcomings stand out.

First, the chord and pitch distributions are rarely tuned for the actual operating Reynolds numbers, so the blades run at a sub-optimal lift-to-drag ratio across most of the span. Second, common blade materials like ABS and basic nylon are soft enough that centrifugal and aerodynamic loading cause significant tip deflection at operating speeds — deflection that reduces effective pitch and therefore thrust. Third, manufacturers seldom check whether blade natural frequencies coincide with motor excitation harmonics, so vibration issues only emerge after the product is in the field. Fourth, the fatigue life of injection-moulded thermoplastics is far shorter than composite alternatives.

This project tackles those gaps head-on by combining BEMT-based analytical design, ANSYS CFD simulation, and a structured material comparison into a single coherent workflow — one that is transferable to future propeller design studies.

1.3 Objectives

The specific objectives are as follows:

- Justify the selection of the Eppler E63 airfoil and document its aerodynamic characteristics at the relevant Reynolds numbers.
- Carry out a complete BEMT-based propeller design, including station-by-station chord and pitch distributions, and non-dimensional performance coefficients.
- Build a 3D CAD model in ANSYS SpaceClaim and set up the computational domain for CFD.
- Generate a high-quality polyhedral mesh with prismatic boundary layer inflation using ANSYS Fluent Meshing.
- Run steady-state RANS CFD simulations at four angles of attack (0°, 4°, 8°, 12°) using the Realizable k-epsilon model.

- Extract and compare velocity contours, pressure distributions, aerodynamic coefficients, and mass flow rates from CFD.
- Validate CFD results against BEMT predictions and published experimental data.
- Evaluate the structural performance of CFRP, PLA, and Nylon under operating loads.
- Draw practical conclusions and recommendations for quadcopter propeller design.

1.4 Scope of Work

This study covers the aerodynamic and preliminary structural analysis of a single two-bladed propeller operating in isolation. Rotor-to-rotor interactions between the four motors of the quadcopter are not modelled. Operating conditions include hover (zero forward speed) and cruise at 16 m/s, at sea-level International Standard Atmosphere conditions. Material comparison considers CFRP, PLA, and Nylon.

1.5 Report Organisation

Chapter 2 reviews relevant published work. Chapter 3 describes the methodology — BEMT design, geometry creation, meshing, and CFD setup. Chapter 4 presents the detailed design and simulation inputs. Chapter 5 covers all results and their discussion. Chapter 6 states conclusions and directions for future work.

CHAPTER 2

LITERATURE REVIEW

2.1 UAV Propeller Aerodynamics — Overview

Research on small UAV propellers has accelerated sharply since about 2015, driven by the rapid commercialisation of consumer drones. Early work by Chua et al. (2017) established basic thrust equations for multirotor configurations and showed how thrust-to-weight ratio requirements translate into rotor disc loading. Eraslan et al. (2020) reviewed methods for determining propeller thrust and power coefficients across a range of UAV scales, noting that BEMT-based approaches consistently outperform simple momentum theory when validated against measurement. Sapit (2021) used CFD to analyse a commercial drone propeller and found that standard RANS turbulence models can predict thrust to within roughly 10% of measured values — an acceptable level of accuracy for preliminary design.

2.2 Low-Reynolds-Number Airfoil Research

The aerodynamic peculiarities of the low-Re regime have been studied extensively since the 1980s. Eppler's own book (1990) remains the definitive reference on airfoil design for this regime, and the UIUC Airfoil Database continues to be the most-used source of experimental and computed polars for profiles like the E63. Work compiled at Chalmers University (Crona et al., 2024) compared measured performance of several low-Re propellers and highlighted how sensitive the aerodynamic efficiency is to even small changes in Reynolds number — reinforcing the need to use a profile explicitly designed for the operating range.

2.3 Blade Element Momentum Theory Studies

BEMT has been the workhorse of propeller design for decades. Hasnaoui et al. (2024) applied BEMT to a small drone rotor and showed that predictions lie within 8% of CFD results in the pre-stall regime, degrading to roughly 15% near stall because the theory

does not model flow separation. Jiménez et al. (2022) conducted an exhaustive comparison of BEMT implementations and found that including Prandtl tip-loss correction is essential for two-bladed rotors at low advance ratios — without it, hover thrust is overestimated by 10–15%. Ganesan and Jayarajan (2022) specifically studied the E63 propeller using BEMT and CFD and reported good mutual agreement for the outer 70% of the blade span, where most of the thrust is generated.

2.4 CFD Methods for Rotating Rotors

The Multiple Reference Frame (MRF) approach has become the standard choice for steady-state rotor simulations because it avoids the computational cost of time-accurate sliding mesh calculations while capturing the mean flow field accurately. Elnady and Hamada (2022) demonstrated MRF and sliding mesh for a full quadcopter in hover and showed that MRF predictions of per-rotor thrust agreed with sliding mesh to within 5%. Christodoulou et al. (2019) confirmed similar accuracy for isolated-rotor CFD across a range of operating conditions. The choice of turbulence model matters: Slingluff (2016) found that the Realizable k-epsilon model with Enhanced Wall Treatment handles the transitional boundary layers present on low-Re blades more reliably than the standard k-epsilon formulation.

2.5 Material Studies on UAV Propellers

The structural side of propeller design has received increasing attention as carbon-fibre propellers have become more affordable. Ahmad et al. (2019) compared CFRP and PLA blades under centrifugal and aerodynamic loading, finding that CFRP achieved tip deflections more than seven times smaller than PLA for the same geometry. Ruseno (2018) studied the modal response of CFRP rotors and noted that the higher stiffness of composites shifts natural frequencies well above motor excitation harmonics, avoiding resonance. More recently, Varol et al. (2025) and Mustafa (2025) confirmed that polymer propellers operating at high RPM carry safety factors well below what would be acceptable for sustained flight cycles.

Taken together, the literature points clearly in the direction followed by this project: use a purpose-designed low-Re airfoil, validate BEMT with CFD, and make CFRP the preferred material wherever structural performance is a priority.

CHAPTER 3

METHODOLOGY USED

3.1 Overall Design Approach

The workflow follows a logical progression from analytical design to numerical validation. BEMT provides the initial propeller geometry — chord and twist distributions that maximise aerodynamic efficiency at the design operating point. Those distributions are then realised as a 3D CAD model in ANSYS SpaceClaim. The CAD geometry is exported to ANSYS Fluent Meshing, where a polyhedral volume mesh is generated with refined boundary layer cells near the blade surface. The mesh is then loaded into ANSYS Fluent for RANS CFD simulation. Results from CFD are compared with BEMT predictions and published experimental data to assess accuracy.

Structural behaviour is evaluated through a one-way coupling approach: aerodynamic pressure loads from the converged CFD solution are mapped onto a finite element model, and tip deflection and Von Mises stresses are extracted for each material. This stops short of full aeroelastic coupling but gives a reliable first estimate of which materials are structurally viable.

3.2 Geometry Preparation

Airfoil coordinates for the E63 profile were downloaded from the UIUC Airfoil Database in Selig format. In ANSYS SpaceClaim, the coordinates were imported and scaled to the chord length appropriate for each of nine spanwise stations (r/R = 0.20 to 1.00). Each cross-section was rotated to the local blade pitch angle computed by BEMT. The Skin/Loft tool then blended all nine profiles into a smooth, continuous blade surface. A cylindrical hub of 30 mm diameter and 40 mm height was Boolean-united with the blade, and the second blade was created by a 180° rotational copy.

The computational domain was built around the Multiple Reference Frame (MRF) philosophy. A rotating inner cylinder — 1.5D diameter, 1.2D height, centred on the hub — contains the propeller and rotates at 6400 RPM in the simulation. A stationary

rectangular outer domain (8D upstream, 8D downstream, 6D lateral) surrounds it. Named selections for blade, hub, inlet, outlet, lateral walls, and the rotor-stator interface were assigned before export.

3.3 Meshing Strategy

Meshing was performed entirely in ANSYS Fluent Meshing 2023 R1 using the Watertight Geometry workflow. Local surface mesh refinement was applied aggressively to the blade surfaces (0.8 mm target size) and progressively coarser away from the blade, reaching 50 mm in the far field. The boundary layer is resolved with 12 prismatic layers, first-cell height set to achieve y^+ = 1–5 on the blade surface, and growth rate of 1.2 — giving total inflation thickness of roughly 0.40 mm. The prismatic core is then converted to polyhedral volume cells, which reduces total cell count compared to a tetrahedral mesh of similar accuracy.

After volume mesh generation, a quality improvement pass was run to reduce maximum skewness below 0.85. The final mesh contained approximately 1.8 million cells, with orthogonal quality above 0.22 and maximum aspect ratio of 14.2 — all within acceptable bounds for this class of problem.

3.4 CFD Simulation Setup

All simulations used the pressure-based SIMPLE solver in steady-state mode, appropriate because the blade tip Mach number is only 0.244. The Realizable k-epsilon turbulence model with Enhanced Wall Treatment was selected based on its documented reliability for rotating machinery flows and its ability to handle y^+ values that vary across the blade surface. Second-order upwind discretisation was applied to all transport equations. PRESTO! was chosen for the pressure interpolation scheme because of its suitability for swirling and rotating flow domains.

Inlet boundary conditions were set to a velocity of 16 m/s (cruise) or 0 m/s (hover), with turbulence intensity of 0.5% — representing a clean, low-turbulence free stream. The outlet was a pressure outlet at zero gauge pressure. Lateral domain walls were treated as symmetry planes. Blade and hub surfaces are no-slip walls in the rotating frame. Convergence was declared when all residuals fell below 1×10^{-4} and force monitors stabilised within ±0.5% over the final 50 iterations.

CHAPTER 4

DESIGN AND CFD ANALYSIS OF EPPLER E63 UAV PROPELLER

4.1 Quadcopter Specifications and Airfoil Selection

4.1.1 System-Level Design Requirements

The target platform is a 1.2 kg quadcopter with four identical propellers. For safe hover and meaningful manoeuvring margin a thrust-to-weight ratio of 2.0 was adopted, giving a total thrust requirement of 2 × 1.2 × 9.81 = 23.54 N and a per-rotor hover target of 5.886 N. Design top speed was fixed at 16 m/s.

Parameter	Symbol	Value	Unit
Total UAV Mass	m	1.2	kg
Number of Propellers	N	4	—
Design Top Speed	V	16	m/s
Blades per Propeller	B	2	—
Design RPM	n	6400	RPM
Propeller Diameter	D	0.25 (≈10 in)	m
Air Density (ISA SL)	ρ	1.225	kg/m³
Dynamic Viscosity	μ	1.789×10■■	Pa·s
Hover Thrust / Prop	T_hover	5.886	N

Table 2.1 — Quadcopter System Specifications

4.1.2 Why the Eppler E63?

The local Reynolds number at a blade element depends on both the rotational velocity and the forward speed. Across the blade span at cruise conditions (r/R = 0.20 to 0.90), Re runs from roughly 42,800 to 87,100. The entire blade sits in what is sometimes called the low-Re regime. Below about Re = 100,000 a few things happen that make the flow hard to deal with using ordinary airfoils. Laminar separation bubbles tend to form on the upper surface, particularly at moderate angles of attack, and they can trigger a sudden

drag increase. Viscous drag is proportionally heavier than at higher Reynolds numbers because boundary layers are thicker relative to blade chord. And trailing-edge separation appears earlier than you might expect.

The NACA 4412 — a common comparison baseline — has a minimum profile drag coefficient of about 0.018 at Re = 100,000. The E63 brings that down to roughly 0.012. That 33% drag reduction might sound modest in isolation, but across a full blade span and at 6400 RPM it accumulates into a meaningful power saving.

Property	Value
Maximum Thickness	10.4% chord at x/c = 0.243
Maximum Camber	5.9% chord at x/c = 0.525
Designed By	Richard Eppler
Design Re Range	60,000 – 300,000
Zero-Lift Angle	–5.5°
Optimal AoA (Re=100k)	4°
C_L at Design Point	0.95
C_D at Design Point	0.015
Peak C_L/C_D	≈ 63
Maximum C_L	1.15
Stall Angle	≈ 13°

Table 2.2 — Eppler E63 Airfoil Aerodynamic Properties

Parameter	E63	NACA 4412	Advantage
Designed for low Re	Yes	No	E63
Re operating range	60k–300k	200k+	E63
(C_L/C_D)_max @ Re=100k	≈ 63	≈ 45	E63 (+40%)
C_D minimum	0.012	0.018	E63 (–33%)
Figure of Merit	0.80	0.75	E63 (+6.7%)
Hover power / prop	51.5 W	54.9 W	E63 (–6.2%)
Cruise efficiency	80%	75%	E63 (+5 pp)

Table 2.3 — E63 vs NACA 4412 at Re = 100,000

4.2 Propeller Geometry Design

4.2.1 Chord Distribution

A linearly tapering chord was adopted to keep the design simple enough to manufacture while still distributing thrust reasonably across the span. The chord law is:

```
c(r/R) = 0.030 × [1 − 0.5 × (r/R)] (metres)
```

This gives a root chord of 27.0 mm at r/R = 0.20, tapering to 15.0 mm at the tip. The 75%R reference chord — conventionally used to specify pitch — is 18.8 mm.

4.2.2 Blade Pitch Distribution

The blade pitch angle at each station was set so that the local angle of attack equals the E63 design value of 4° at cruise. The inflow angle at each station is $\phi(r) = \arctan(V / V_{rot})$, so the pitch follows $\beta(r) = \phi(r) + 4°$.

r/R	r (m)	V_rot (m/s)	ϕ_cruise (°)	β (°)	Status
0.20	0.0250	16.76	43.69°	47.69°	■ Hub stall
0.30	0.0375	25.13	32.48°	36.48°	■ Stall
0.40	0.0500	33.51	25.57°	29.57°	■ Near stall
0.50	0.0625	41.89	20.93°	24.93°	■ Marginal
0.60	0.0750	50.27	17.67°	21.67°	≈ Limit
0.70	0.0875	58.64	15.26°	19.26°	✔ Safe
0.75★	0.0938	62.83	14.29°	18.29°	✔ Safe
0.80	0.1000	67.02	13.42°	17.42°	✔ Safe
0.90	0.1125	75.40	11.97°	15.97°	✔ Safe
1.00	0.1250	83.78	10.81°	14.81°	✔ Safe

Table 2.4 — Blade Pitch Distribution (Cruise, V = 16 m/s, n = 6400 RPM)

Inner stations stalling in hover is a known and accepted trait of fixed-pitch propellers that are optimised for cruise. The inner 30% of the blade contributes less than 15% of total thrust anyway — the outer stations carry the load.

4.3 BEMT Mathematical Calculations

4.3.1 Core BEMT Equations

For each blade element at radius r with width dr, the following sequence is applied:

```
V_eff = sqrt(V_rot² + V∞²) [Effective resultant velocity]
Re = ρ × V_eff × c(r) / μ [Local Reynolds number]
dT/dr = (dL/dr)·cos(ϕ) − (dD/dr)·sin(ϕ) [Thrust per blade per unit span]
dQ/dr = r × [(dL/dr)·sin(ϕ) + (dD/dr)·cos(ϕ)] [Torque per blade per unit span]
dT_total/dr = B × dT/dr × F_tip [With Prandtl tip-loss correction]
```

4.3.2 E63 Aerodynamic Data vs Reynolds Number

Re Range	C_L at α=4°	C_D at α=4°	C_L/C_D	Note
< 50,000	0.75	0.035	21.4	Very low Re
50k – 65k	0.78–0.82	0.025–0.028	30–33	Transition zone
65k – 80k	0.88–0.92	0.018–0.020	46–51	Improving
80k – 100k	0.94–0.95	0.015–0.016	59–63	Near design point
100k – 130k	0.95	0.015	63	Design point

Table 3.1 — E63 C_L/C_D vs Reynolds Number at α = 4° (Source: XFOIL, Ncrit=9)

4.3.3 Full Station-by-Station Results

r/R	c (m)	V_eff	Re	C_L	C_D	F_tip	dT_total/dr
0.20	0.0270	23.17	42,820	0.75	0.035	1.00	9.20
0.30	0.0255	29.79	52,020	0.78	0.028	1.00	17.84
0.40	0.0240	37.13	61,020	0.88	0.020	1.00	31.82
0.50	0.0225	44.84	69,160	0.90	0.018	1.00	46.26
0.60	0.0210	52.75	75,900	0.92	0.017	1.00	62.40
0.70	0.0195	60.78	81,220	0.94	0.016	1.00	79.74
0.75★	0.0188	64.84	83,250	0.95	0.015	1.00	89.06
0.80	0.0180	68.90	84,940	0.95	0.015	0.85	82.00
0.90	0.0165	77.08	87,120	0.95	0.015	0.60	66.83

Table 3.2 — BEMT Station-by-Station Results (Cruise, V∞ = 16 m/s)

4.3.4 Non-Dimensional Performance Summary

Coefficient	Hover	Cruise (J=0.60)	Unit
Thrust Coefficient C_T	0.1082	0.0541	—
Power Coefficient C_P	0.0354	0.0406	—
Torque Coefficient C_Q	0.0056	0.0065	—
Advance Ratio J	0	0.60	—
Propulsive Efficiency η	—	80%	—
Figure of Merit FM	0.80	—	—
Power per Propeller	51.5	59.0	W
Total System Power (×4)	206.0	236.0	W

Table 3.4 — BEMT Non-Dimensional Performance Summary

4.4 CFD Simulation — Detailed Setup

4.4.1 Meshing Parameters

Metric	Target	Achieved	Status
Total cell count	1–3 million	~1.8 million	✔
Orthogonal quality (min)	≥ 0.1	0.22	✔
Skewness (max)	≤ 0.85	0.78	✔
Aspect ratio (max)	≤ 20	14.2	✔
y■ on blade	1–10	2–8	✔
Inflation layers	12	12	✔
First layer height	0.02 mm	0.02 mm	✔

Table 3.6 — Mesh Quality Assessment

4.4.2 Boundary Conditions and Solver Settings

Table 3.7 summarises boundary conditions. The turbulence model was Realizable k-epsilon with Enhanced Wall Treatment. PRESTO! was used for pressure interpolation; all other transport equations used second-order upwind discretisation. The inner cylinder rotated at 6400 RPM using MRF. Convergence criterion was residuals below 1×10^{-4} with force monitor stabilisation within ±0.5% over the last 50 iterations.

Boundary	Type	Key Setting	Value
Inlet	Velocity Inlet	V (cruise)	16 m/s
Inlet	Velocity Inlet	V (hover)	0 m/s
Outlet	Pressure Outlet	Gauge pressure	0 Pa
Lateral Walls	Symmetry	—	—
Blade Surfaces	Wall (no-slip)	In rotating frame	Stationary
Rotating Zone	Cell Zone	Rotational speed	6400 RPM
Interface	Mesh Interface	Coupling	Sliding

Table 3.7 — Boundary Conditions Summary

Material	Density (kg/m³)	Young's Mod. (GPa)	Poisson Ratio	Tensile Str. (MPa)
CFRP	1600	70	0.30	600
PLA	1240	3.5	0.36	65
Nylon (PA12)	1010	1.6	0.41	50
ABS (ref.)	1050	2.1	0.39	40

Table 5.1 — Material Properties for Structural Analysis

CHAPTER 5

RESULTS AND DISCUSSION

5.1 CFD Convergence

Every simulation in the parametric matrix converged cleanly. All residuals — continuity, three momentum components, turbulent kinetic energy, and dissipation rate — dropped below 1×10^{-4}. The cruise case at 4° AoA reached this level by iteration 420; hover cases took a little longer, typically 500–700 iterations. Thrust and torque monitors settled to within ±0.5% of their final values well before the residuals fully converged. Mass flow imbalance at the outlet was below 0.01% in all cases, confirming that the mesh conserves mass acceptably.

5.2 Velocity Contours and Flow Patterns

At the design angle of attack of 4°, the velocity contours tell the expected story. Flow stagnates just ahead of the leading edge and accelerates strongly over the suction surface, peaking at around 85 m/s near the 75%R station. The wake downstream of the rotor plane shows a clean, roughly uniform velocity deficit — a sign that the blade is operating in attached flow throughout most of its span. The tip vortex rolling at the blade tip is clearly visible in the contours.

At 0° AoA the picture is broadly similar but with a smaller pressure differential between the blade surfaces; peak suction surface velocity drops to around 78 m/s. At 8°, suction surface velocity climbs to about 92 m/s and the first signs of trailing-edge separation appear on the inner 30% of the blade — exactly where the pitch analysis in Section 4.2.2 flagged marginal and near-stall conditions. At 12°, separation is unmistakable. A large, slow-moving region (2–4 m/s) covers 30–40% of the blade area, and the wake becomes markedly more turbulent. The E63 stall angle is around 13°, so this result is physically consistent.

5.3 Pressure Distribution

The static pressure contours on the blade surface at 4° show a clear and well-resolved pressure differential: the pressure surface (underside) carries positive gauge pressure in red-to-orange tones, while the suction surface shows the characteristic blue negative-pressure region. The pressure difference peaks around the 75%R station, which aligns with both the highest chord Reynolds number and the highest rotational velocity for meaningful lift generation.

Pressure coefficient distributions at the 75%R cross-section confirm the expected behaviour. The suction peak — the sharp minimum in C_p on the upper surface — sits at about 10% chord at 4° AoA and migrates forward to about 5% chord at 12°, the classic signature of an airfoil approaching stall. The suction peak magnitude grows from roughly −3.2 at 4° to −5.8 at 12°, reflecting the increasing angle of attack.

5.4 Aerodynamic Coefficients

Table 5.1 sets out the lift and drag coefficients from CFD alongside BEMT analytical predictions and published experimental data for the E63 at Re = 80,000–100,000.

AoA (°)	C_L (CFD)	C_L (BEMT)	C_L (Exp.)	C_D (CFD)	C_D (BEMT)	C_D (Exp.)	C_L/C_D (CFD)	C_L/C_D (BEMT)
0°	0.41	0.39	0.39	0.0165	0.0165	0.016	24.8	23.6
4°	0.93	0.95	0.94	0.0162	0.0150	0.015	57.4	63.3
8°	1.08	1.10	1.09	0.0235	0.0210	0.022	45.9	52.4
12°	1.10	1.12	1.10	0.0480	0.0420	0.045	22.9	26.7

Table 5.1 — Aerodynamic Coefficients: CFD vs BEMT vs Experimental

A few things are worth pointing out here. The CFD lift coefficient at 4° is 0.93 versus the experimental 0.94 — a 1.1% difference that is essentially negligible. The drag coefficient at the design point is slightly overestimated by CFD (0.0162 versus 0.015 from both BEMT and experiment), giving a CFD C_L/C_D peak of 57.4 against the experimental 62.7. That 8% drag overprediction is consistent with known tendencies of RANS models in transitional boundary layers and was expected going in. Beyond 8° AoA, the divergence between CFD and BEMT grows, which makes sense — BEMT cannot model separation, so it continues to predict optimistically where the flow has already broken down.

5.5 Mass Flow Rate Results

AoA (°)	Mass Flow — Outlet (kg/s)	Mass Flow — 1D Downstream (kg/s)	Thrust from CFD (N)
0°	3.421	3.415	2.12
4°	3.865	3.858	4.32
8°	4.102	4.098	5.02
12°	4.215	4.210	4.95

Table 5.2 — Mass Flow Rate and Thrust at Various AoA

Mass flow through the rotor plane increases monotonically with AoA, as expected — higher angle of attack means more work done on the fluid. The slight drop in thrust between 8° and 12° despite continued mass flow increase is the drag penalty of incipient stall showing up in the force balance.

5.6 Thrust and Power Analysis

AoA (°)	Thrust-CFD (N)	Thrust-BEMT (N)	Torque-CFD (N·m)	Power-CFD (W)	C_T	C_P	η_p
0°	2.12	2.05	0.068	45.6	0.0389	0.0314	74.1%
4°	4.32	4.48	0.118	79.1	0.0794	0.0544	87.6%
8°	5.02	5.20	0.156	104.6	0.0922	0.0720	76.6%
12°	4.95	5.05	0.195	130.7	0.0910	0.0899	60.6%

Table 5.3 — Thrust, Torque, and Power vs AoA from CFD

The hover target of 5.886 N per propeller is not quite reached at the design cruise RPM of 6400 in these simulations, which use the cruise pitch setting. In actual flight the flight controller would increase RPM to around 5,400–5,800 RPM in hover to hit the required thrust. That is normal and expected for a fixed-pitch propeller designed around a cruise operating point. Peak propulsive efficiency of 87.6% at 4° AoA represents an impressive result for a rotor of this scale and confirms that the E63 design point was correctly identified.

5.7 Structural Material Comparison

The one-way FSI analysis told a fairly clear story about material choice. CFRP stands in its own category.

Material	Density (kg/m³)	Tip Deflection (mm)	Von Mises Stress (MPa)	Strength (MPa)	Safety Factor	Status
CFRP	1600	0.28	42.5	600	14.1	✔ Excellent
Nylon (PA12)	1010	1.42	38.8	50	1.3	■ Marginal
PLA	1240	2.14	52.1	65	1.2	■ Marginal
ABS (ref.)	1050	2.88	58.3	40	0.7	✘ Fails

Table 5.4 — Structural Material Comparison at 6400 RPM

The 2.14 mm tip deflection of PLA is not just an abstract number. At that magnitude, the effective pitch angle at the tip drops by about 0.9°, which translates to a thrust loss in the range of 3–4%. Over the lifetime of the battery pack, that is a measurable reduction in endurance. For Nylon the situation is better but still marginal, with a safety factor of only 1.3 under static loading — a value that provides very little reserve against fatigue under repeated loading cycles. ABS, the most common commercial propeller material, fails the static strength check outright. CFRP, with a safety factor above 14 and tip deflection under 0.3 mm, is the only material that justifies confident use in a performance-critical application.

CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusions

This study has demonstrated a complete, end-to-end analytical and numerical design process for a small UAV propeller — from initial sizing through BEMT, to CFD validation and structural material screening. Several concrete conclusions emerge from the work.

The Eppler E63 is the right choice for this application. At the operating Reynolds numbers (Re = 42,000–87,000), it delivers a peak C_L/C_D of roughly 57–63 depending on the method used, which is 40% better than NACA 4412 at the same conditions. That performance advantage translates into 6.2% lower hover power, a direct benefit to battery endurance.

BEMT, when used with Prandtl tip-loss correction and span-varying aerodynamic data, predicts thrust and efficiency to within 8–12% of CFD in the normal operating range. Beyond 8° AoA — where separation begins — BEMT overpredicts thrust by up to 15%, consistent with its inability to model separated flow. For preliminary design purposes BEMT remains highly useful; for near-stall or off-design conditions CFD is essential.

The polyhedral mesh with prismatic boundary layers, combined with the Realizable k-epsilon turbulence model and Enhanced Wall Treatment, produced well-converged and physically consistent results across all four AoA cases. CFD is somewhat conservative on C_L/C_D (the drag is slightly overestimated) but this is a known and conservative bias that does not affect design decisions.

On the structural side, CFRP is the recommended material with no real qualification. A safety factor above 14 and tip deflection well below 0.3 mm at 6400 RPM sets it apart from every polymer alternative tested. PLA and Nylon are marginal; ABS fails. Any propeller that needs to hold its pitch under aerodynamic and centrifugal

load — which means every performance-critical application — should be made from CFRP.

6.2 Future Scope

Several extensions of this work would be valuable. The most impactful would be replacing the steady-state MRF simulation with a time-accurate sliding mesh transient simulation, which would capture the unsteady wake structure, blade-vortex interaction, and rotor-to-rotor effects in a full quadcopter configuration. That is computationally more expensive but now tractable on modern multi-core workstations.

The BEMT calculation used a single representative Reynolds number for each blade station. Extending it to use Re-dependent polars from XFOIL or wind tunnel measurements, particularly for the inner blade stations where Re drops below 50,000, would improve hover thrust prediction.

A two-way fluid-structure interaction simulation coupling ANSYS Fluent with ANSYS Mechanical would quantify the aeroelastic thrust reduction under operating loads for polymer blades — the one-way approach used here cannot capture the feedback between blade deflection and aerodynamic loading.

Finally, and perhaps most importantly, physical testing. Fabricating CFRP and PLA propellers and running them on a thrust stand at the ABES Engineering College laboratory would provide the definitive validation of both BEMT and CFD predictions. Publishing the measured data would also contribute to the still-sparse body of experimental data on E63 propellers at this scale.

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